

ANALYSIS REPORT

Trader Performance Analysis using Bitcoin Fear & Greed Index

Objective

The objective of this project was to analyze cryptocurrency trader performance under different market sentiment conditions by combining historical trading records with the Bitcoin Fear & Greed Index. The analysis aimed to identify behavioral patterns, evaluate profitability, and provide actionable recommendations.

Methodology

- Loaded and cleaned both datasets.
- Converted timestamps into a common date format.
- Merged trading data with sentiment data.
- Engineered features including trade outcome and daily PnL.
- Performed exploratory data analysis using statistical summaries and visualizations.
- Compared trader performance across different market sentiment categories.

Key Findings

- The highest trader win rate (46.49%) occurred during Extreme Greed.
- Fear periods recorded the largest trading activity with 61,837 trades.
- Average trade size peaked at \$7,816 during Fear conditions.
- Larger trading volume and position size did not necessarily result in better trading performance.

Recommendations

- Integrate market sentiment into trading strategies.

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- Focus on disciplined trade selection rather than increasing trading frequency.
- Adjust position size according to market volatility and sentiment.

Conclusion

The analysis demonstrates that market sentiment has a measurable impact on trader performance and behavior. Incorporating sentiment indicators into trading decisions can improve risk management and support more informed trading strategies.